Suyzeko®

Suyzeko hydrogen inhalation machine

Our hydrogen inhalers adopt **PEM/SPE Electrolysis** Technology and are equipped with **DuPont N117 proton exchange membrane.** This allows for efficient decomposition to produce hydrogen and oxygen, with the hydrogen-oxygen ratio strictly controlled at 2:1. The hydrogen purity reaches **99.996%**, ensuring the purity of the gas.

*Warranty: 3 years
*Medical grade feature: Medical standard circuit design and components
*Test report from professional lab: CE, EMC, LVD, DOC, FCC, UL, PSE,ROHS certificates
*Multi therapy options: Pair with hydrogen water drinking, eye therapy, ear therapy(ask us about bundle deals or how to combine the different therapy programs).

The core function of the hydrogen inhalation machine: By providing high-purity hydrogen gas, it utilizes the selective antioxidant and anti-inflammatory properties of hydrogen molecules to comprehensively improve the health condition of the human body.

Main applications: Anti-aging and antioxidant, improvement of respiratory system diseases, enhancement of sleep quality, relief of fatigue, assistance in the management of chronic diseases, and promotion of post-illness recovery.

This is our main model of hydrogen inhalation machine. The machine flow ranges from 150 to 6000 ml/min:

Hydrogen Oxygen generator

GY-H150

GY-H200/300

GY-225/300/450

GY-H600

GY-HX900

GY-HX1500

GY-HX1800

GY-HX3000

GY-HX3600

GY-HX6000

900ml/min

450ml/min

1500ml/min

150ml/min

3000ml/min

600ml/min

3600ml/min

3000ml/min

600ml/min

900/1500ml/min

200/300ml/min

900ml/min

1200ml/min

3600ml/min

3000ml/min

300/450ml/min

What is SPE Electrolysis Technology? What are the advantages of our hydrogen inhalation machine using SPE Electrolysis Technology + DuPont N119 Proton Membrane?

I. What is SPE Electrolysis Technology?

SPE (Solid Polymer Electrolyte) electrolysis technology is a high-end, safe pure water electrolysis hydrogen production technology, whose core is to replace the liquid electrolyte (such as KOH solution) in traditional alkaline electrolysis with a proton exchange membrane (PEM).

Core Principles:

1. The electrolyzer is divided into two independent chambers—the anode chamber and the cathode chamber—by the proton membrane.
2. Water is electrolyzed at the anode:

$$2H_2O \rightarrow O_2 \uparrow + 4H^+ + 4e^-$$

producing oxygen, protons (H^+), and electrons.

3. Protons (H^+) pass through the proton membrane to the cathode, and electrons reach the cathode through an external circuit, combining to form pure hydrogen at the cathode:

$$4H^+ + 4e^- \rightarrow 2H_2 \uparrow$$

4. Finally, complete separation of hydrogen and oxygen is achieved, with pure hydrogen and pure oxygen collected separately.

Core Advantages of SPE Technology

(for Household Hydrogen Inhalation Machines)

***Produces pure hydrogen**: Hydrogen purity is usually ≥99.99%, with no oxygen or impurities mixed in, making inhalation safer.

***High safety**: Separation of hydrogen and oxygen eliminates the explosion risk of mixed gas; no liquid electrolyte means no corrosion or leakage hazards.

***High efficiency**: Higher electrolysis efficiency than traditional alkaline electrolysis, fast startup, low energy consumption, suitable for frequent start-stop in household use.

***Simple maintenance**: Only deionized water is needed, no need to replenish liquid electrolyte, making the equipment cleaner and more durable.

II. Benefits of Using DuPont N119 Proton Membrane

DuPont N119 belongs to the Nafion series of perfluorosulfonic acid proton membranes, which is recognized as the "gold standard" for high-end proton membranes in the industry, and is especially suitable for household hydrogen inhalation machines. Its core advantages are as follows:

1. Ultra-high proton conductivity → Faster hydrogen production, more energy-saving

N119 membrane has excellent proton conductivity, enabling more efficient electrolysis reactions:

- Faster hydrogen production at the same power;
- Lower energy consumption for the same hydrogen output, more economical for long-term use.

2. Extreme chemical stability → Long service life, less maintenance

The perfluorosulfonic acid structure gives N119 membrane:

*Resistance to acids, alkalis, oxidation, and hydrolysis, not easily degraded or aged in the strong oxidation/reduction environment of electrolysis for a long time;

*In household hydrogen inhalation machines, the membrane service life can usually reach 2–3 years or even longer, greatly reducing replacement frequency and maintenance costs.

3. Excellent mechanical strength → Stable operation, low failure rate

N119 is a thick membrane (thickness approx. 127μm) with high mechanical strength:

*Resistant to stretching and tearing, able to withstand pressure and temperature fluctuations during electrolysis;

*Not easy to break or leak, making the equipment operate more stably with a lower failure rate.

4. Perfect gas selectivity → Maximized hydrogen purity

The membrane has extremely strong selectivity, only allowing protons to pass through, effectively blocking gas cross-permeation:

*Ensures hydrogen purity ≥99.99%, with no oxygen or impurities mixed in;

*Safer for inhalation, with more stable and reliable hydrogen-rich effects.

5. Industry benchmark quality → "Standard configuration" for high-end equipment

DuPont Nafion series is the industry benchmark for proton membranes, and N119, as a thick and durable model, is the preferred membrane material for high-end hydrogen inhalation machines and medical-grade electrolyzed water equipment:

*Quality is guaranteed, supporting long-term stable operation of the equipment;

*It is one of the core hardware indicators to distinguish "high-end hydrogen inhalation machines" from "ordinary models".

III. Comprehensive Advantages of SPE + N119 Membrane

***Safety**: Pure hydrogen production + hydrogen-oxygen separation, no explosion risk of mixed gas, no liquid electrolyte leakage.

***Efficiency**: Fast hydrogen production, low energy consumption, more worry-free for daily use.

***Durability**: Long membrane service life, less maintenance, low long-term use cost.

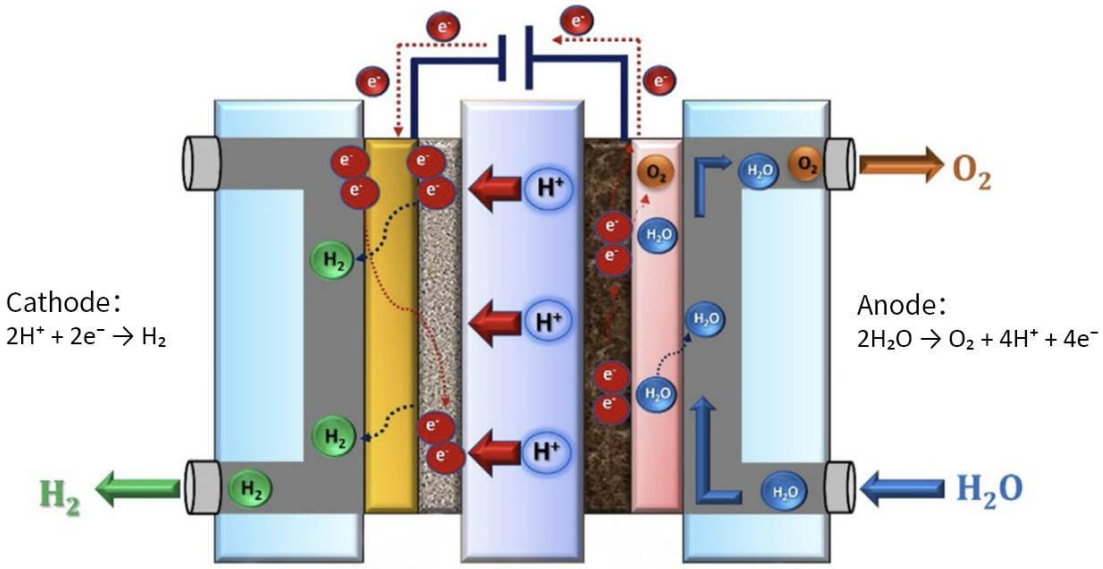
***Purity**: High hydrogen purity, no impurities, healthier for inhalation.

***Stability**: Stable operation, adaptable to different environments, suitable for long-term and high-frequency household use.

A hydrogen inhalation machine using SPE technology + DuPont N119 membrane is a high-end configuration, which is far superior to traditional Brown's gas machines and ordinary proton membrane equipment in terms of safety, efficiency, durability, and hydrogen purity.

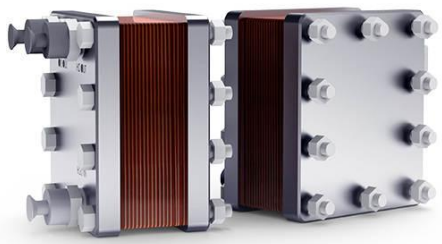
Principle of proton exchange membrane (PEM) electrolysis technology

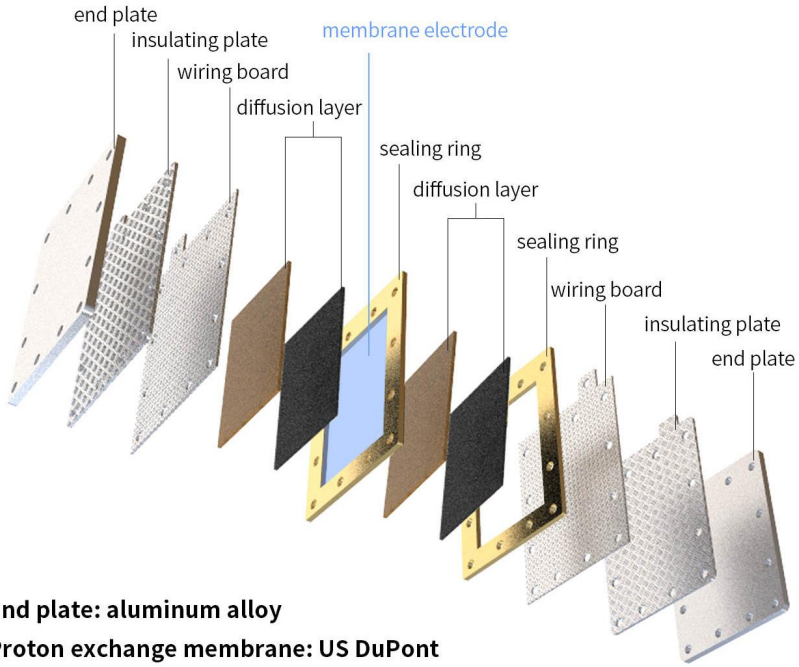
During PEM electrolysis, water is electrochemically decomposed into hydrogen and oxygen. In this process, water molecules first break down on the anode side, producing oxygen (02), protons (H^+), and electrons (e^-). The resulting oxygen is discharged from the anode surface, the remaining protons reach the cathode side through the proton conductive film, and the electrons reach the cathode side through the external circuit. On the cathode side, protons and electrons recombine to produce hydrogen.



Electrolysis cell structure diagram

The main components of the PEM electrolyzer are the membrane electrode assembly (composed of membrane and anode and cathode materials), the gas diffusion layer, the separation plate (bipolar plate) and the end plate.





End plate: aluminum alloy
Proton exchange membrane: US DuPont
Frame board: PPSU
Gasket: PTFE
Wiring board: pure titanium
Diffusion layer: pure titanium
Catalyst: anode loaded iridium, cathode loaded platinum

Our electrolyzer:

• Proton exchange membrane: High performance perfluorosulfonic acid membrane (such as DuPont Nafion or similar high-end membrane) with high proton conductivity, low gas permeability, excellent chemical stability and long life span.

• Catalyst: The use of precious metals (such as iridium, platinum) as a catalyst, high activity, strong stability, significantly improve the efficiency of electrolysis.

• Bipolar plate: titanium + precious metal platinum coating, strong corrosion resistance, reduce contact resistance, extend service life.

• Process technology: Precision manufacturing (such as nanoscale catalyst dispersion technology, zero-pitch membrane electrode components, etc.), ensure efficient reaction status.

Other electrolyzers:

· Proton exchange membrane: may use non-perfluorinated membrane or domestic low-cost alternative with low proton conductivity, high gas permeability and short life span.

· Catalyst: Reducing the amount of precious metals (such as doping non-precious metals or using low load catalysts), resulting in decreased activity and decay too fast.

· Bipolar plate: stainless steel or graphite-based material, not surface treatment, easy to corrode, high resistance.

· Process technology: The manufacturing process is simple, the membrane electrode catalyst layer is too thick or too thin, the catalyst distribution is uneven, and the reaction efficiency is low.

Suyzeko's hydrogen inhalation products integrate more than 36 global patents, achieving an independent intellectual property from industrial design to core control systems, providing partners with one-stop solutions

1. Aesthetics and Functionality: Exclusive appearance patent
2. Smart Control: Circuit control patent, ensuring stable hydrogen purity of 299.99% and flow precision error of less than ±2%.
3.Modular Design: Replaceable hydrogen-oxygen separation module patent, supporting rapid customization of different flow specifications (150-6000 mL/min) to meet diverse application needs.

Electrolyzer with multi-layer of Iridium (Ir) coating and dupont 117 proton membrane. Factory standard of electrolyzer is 1.8V/pcs. Machine have 3V overvoltage protection, voltage increasing after machine long-term working, but do not have any harmful ozone gas. After 8000 hours electrolyzer aging, hydrogen production decay of less than 10%

1. Thermal stability: lower operating temperature
2. Chemical stability: stronger corrosion resistance
3. Mechanical property: longer service life
4. Performance: more stable, sufficient, higher concentration

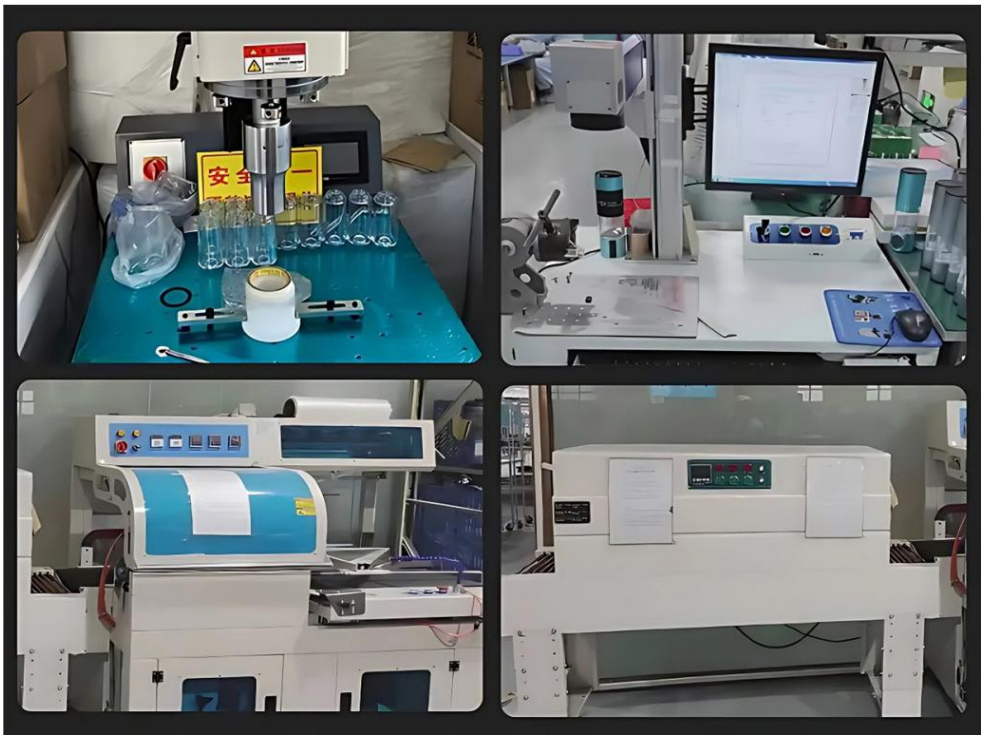
Honor Wall

未检验
未检验

Whole-Process Quality Control System

For the appearance and function of incoming materials, semi-finished parts or finished machines, a total of 30 processes have 100% data recording and traceability, including raw material batch source, calibration verification of key components, complete test history with operator signature.

1. Electrolyzer aging test: Full load testing for 72 consecutive hours to verify the operational stability.
2. Aging test of semi-finished product: 48 hours of testing to ensure the accuracy of functions and parameters.
3. Complete inspection of finished product: re-verification of the function, effective avoidance of bad appearance, to ensure that the machines meet the quality control.



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